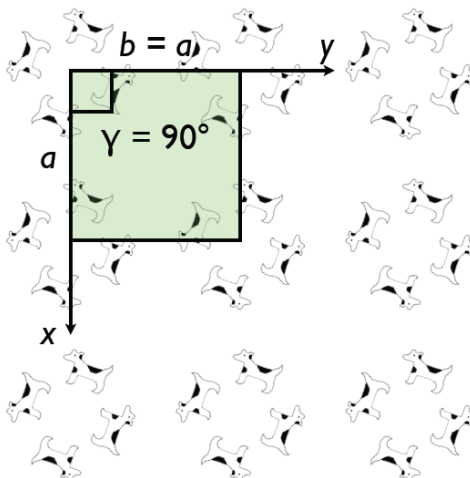


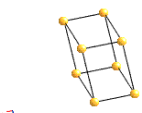
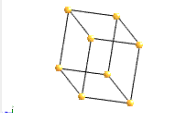
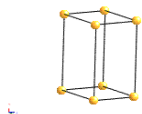
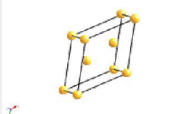
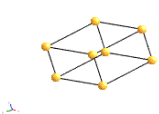
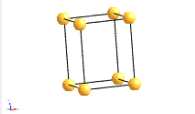
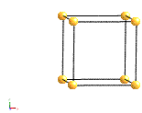
1.7.12: Crystal Systems

The rotational symmetry of a crystal places constraints on the shape of the conventional unit cell we choose to describe the structure. On this basis we divide all structures into one of 7 crystal systems. For example, for crystals with 4 fold symmetry it will always be possible to choose a unit cell that has a square base with $a = b$ and $\gamma = 90^\circ$:

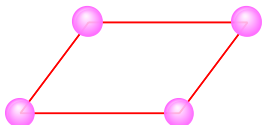


There are 14 unique combinations of the 7 crystal systems with the possible types of primitive and non-primitive lattices. These are referred to as the 14 Bravais lattices.

Crystal systems, lattices and symmetry elements

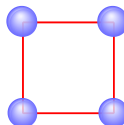
Crystal System	Defining Symmetry	Unit Cell Geometry	
Triclinic	Translational Only	$a \neq b \neq c; \alpha \neq \beta \neq \gamma$	
Monoclinic	A diad axis (parallel to $[010]$)	$a \neq b \neq c; \alpha = \gamma = 90^\circ; \beta > 90^\circ$	
Orthorhombic	3 diads (each should be parallel to each axis)	$a \neq b \neq c; \alpha = \beta = \gamma = 90^\circ$	
Trigonal For more information click here	1 triad (parallel to $[001]$)	$a = b \neq c; \alpha = \beta = 90^\circ; \gamma = 120^\circ$ (or $a = b = c; 120^\circ > \alpha = \beta = \gamma \neq 90^\circ$)	
Hexagonal	1 hexad (parallel to $[001]$)	$a = b \neq c; \alpha = \beta = 90^\circ; \gamma = 120^\circ$	
Tetragonal	One tetrad (parallel to the $[001]$ vector)	$a = b \neq c; \alpha = \beta = \gamma = 90^\circ$	
Cubic	4 triads (all parallel to $\langle 111 \rangle$ directions)	$a = b = c; \alpha = \beta = \gamma = 90^\circ$	

Primitive triclinic



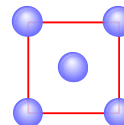
www.doitpoms.ac.uk

Primitive monoclinic



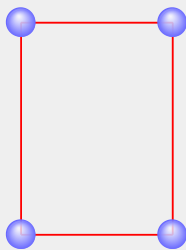
www.doitpoms.ac.uk

C-centered monoclinic



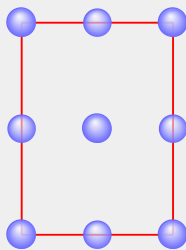
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Primitive orthorhombic



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Face centred orthorhombic



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Please try again later.

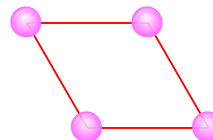
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Service Unavailable

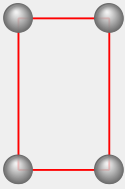
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Hexagonal



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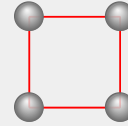
Primitive tetragonal

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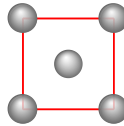
Primitive cubic

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Service Unavailable

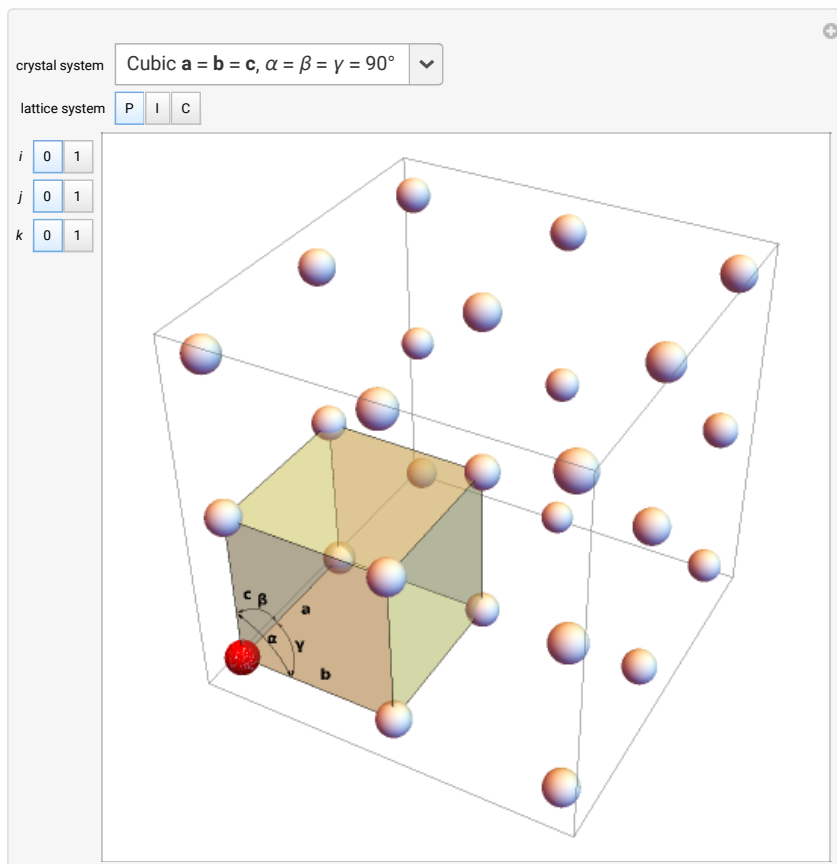
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Please try again later.

Body centered cubic

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And you can use the Wolfram Demonstration Project Viewer to look at the Bravais lattices

The 14 3D Bravais Lattices



This Demonstration shows the characteristics of 3D Bravais lattices arranged according to seven crystal systems: cubic, tetragonal, orthorhombic, monoclinic, triclinic, rhombohedral and hexagonal. Each crystal system can be further associated with between one and four lattices by adding to the primitive cell (click "P"): a point in the center of the cell volume (click "I"), a point at the center of each face (click "F") or a point just at the center of the base faces (click "C"). The points located at the center/faces are highlighted in blue; each point is also a vertex or center of the cell/face, therefore each point is equivalent to every other point.

Crystal systems are determined by the relative lengths of the basis vectors a , b , c and the angles (α , β , γ) between them [1].

It is possible to shift the cell by one unit along a basis vector by selecting the i , j , k values. When repeated, this can generate the entire lattice.